# AICyberAuditBox

## AZURE VM SETUP & INSTALLATION GUIDE

Deploying a Local, Zero-Dependency Offline Compliance Engine

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| DEPLOYMENT TARGET | Microsoft Azure Cloud (IaaS Virtual Machine) |
| OS ARCHITECTURE | Windows Server 2022 / Windows 10 Pro Pro |
| ACTIVE BACKEND | llama.cpp (llama-server.exe) CPU-optimized |
| LOCAL DATABASE | PostgreSQL Master-Slave Docker Containers |
| WEB INTERFACE | Streamlit (Port 8501 bound to 0.0.0.0) |
| AUTO-START TRIGGER | Windows Task Scheduler (On Windows Logon) |
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## 1. Introduction & Core Requirements

This installation guide outlines the exact, step-by-step procedure to set up the AICyberAuditBox compliance auditing application on an Azure Virtual Machine. To ensure compliance with strict privacy guidelines, all auditing computations (document ingestion, paragraph splitting, vector embeddings, and LLM text generation) happen completely locally within the VM virtualized space. No external API calls are made.

**System Specifications Requirement:**

- CPU: Minimum 8 physical cores (optimal cores matching thread settings).
- Memory: 16GB RAM minimum (required to load llama.cpp models and PostgreSQL replication DB).
- Storage: 50GB SSD space (for GGUF model files and Docker volumes).
- Docker: Docker Desktop (configured with WSL2 backend).

## 2. Azure VM Provisioning & Security Group Settings

When creating the VM in the Microsoft Azure Portal, select a Windows Server 2022 Datacenter or Windows 10 Pro image. During provisioning, you must allow RDP (port 3389) and explicitly allow inbound HTTP connections to the Streamlit dashboard.

### Step 1.1: Add Inbound Rule in Azure Portal (Network Security Group)

To access the dashboard from your personal laptop without connecting via RDP:

1. In the Azure Portal, open your Virtual Machine page.
2. Navigate to Networking -> Network settings (left sidebar).
3. Click Add inbound port rule and enter the following settings:

- **Source: Any**
- **Source Port Ranges: ***
- **Destination: Any**
- **Destination Port Ranges: 8501**
- **Protocol: TCP**
- **Action: Allow**
- **Priority: 1000**
- **Name: Allow_Streamlit**

## 3. Repository Deployment & Model Ingestion

Log in to the VM via Remote Desktop (RDP). Open PowerShell and configure the code base and offline language models. The project contains pre-compiled binaries of llama-server.exe and expects offline models to be placed in the project root.

### Step 2.1: Clone the Branch & Verify Update

Navigate to the desktop directory (or your preferred root directory) and execute:

```
  cd C:\Users\veeresh988V\Desktop\llama
git clone -b kv-cache https://github.com/AISecurityComplianceAuditOps/AICyberSecurityAuditBoxV.git
cd AICyberSecurityAuditBoxV
```

### Step 2.2: Acquire Model GGUFs

Copy or download the required GGUF files to the root of the project directory (C:\Users\veeresh988V\Desktop\llama\AICyberSecurityAuditBoxV\):

- LLM Model: google_gemma-4-E4B-it-Q4_K_M.gguf (approx. 2.6GB)
- Embedding Model: nomic-embed-text-v1.5.f16.gguf (approx. 270MB)

### Step 2.3: Start Docker Daemon

Open Docker Desktop inside the VM. Ensure that WSL2 integration is fully active and that the Docker service is running (green status icon).

## 4. Automated Startup Configuration (Auto-Logon & Task)

To ensure that the auditing application automatically launches whenever the Azure VM is started (either scheduled or powered on via mobile), we configure Windows Auto-Logon and a logon task. Auto-Logon triggers a background user session, launching Docker Desktop and the pipeline automatically without requiring manual RDP connection.

### Step 3.1: Configure Windows Auto-Logon (PowerShell)

Run this block in PowerShell as Administrator inside the VM. Replace the password:

```
  Set-ItemProperty -Path "HKLM:\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Winlogon" -Name "AutoAdminLogon" -
Set-ItemProperty -Path "HKLM:\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Winlogon" -Name "DefaultUserName" -V
Set-ItemProperty -Path "HKLM:\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Winlogon" -Name "DefaultPassword" -V
```

### Step 3.2: Register the Windows Task Scheduler Task

Create the scheduled task which executes at logon (automatically triggered by Auto-Logon):

```
  # Define path parameters
$projectDir = "C:\Users\veeresh988V\Desktop\llama\AICyberSecurityAuditBoxV"
$batPath = "$projectDir\run_llamacpp_demo.bat"

# Define execution action
$action = New-ScheduledTaskAction -Execute "cmd.exe" -Argument "/c `"$batPath`"" -WorkingDirectory $projectDir

# Define trigger (runs automatically as the user logs on)
$trigger = New-ScheduledTaskTrigger -AtLogon

# Configure task limits and robustness
$settings = New-ScheduledTaskSettingsSet -AllowStartIfOnBatteries -DontStopIfGoingOnBatteries -ExecutionTimeLim

# Register task with administrative privileges
Register-ScheduledTask -TaskName "AICyberAuditBox_Startup" -Action $action -Trigger $trigger -Settings $setting
```

## 5. Application Verification & Maintenance

Once configured, you can power off the VM to test the automated startup logic. Ensure the server initializes correctly and is reachable externally.

### Step 4.1: Verify Startup Logic

1. Restart your Azure VM from the portal (or from the Azure mobile app).
2. Do not log in using RDP.
3. Wait approximately 60 seconds (giving Windows time to load the model).
4. Open a browser on your laptop (or any other device) and go to:
   **http://40.81.235.37:8501**

### Step 4.2: Manual Startup & Shutdown (For Maintenance)

If you are logged into the VM via RDP, you can control the engine manually using PowerShell:
**Manual Start:**

```
Start-ScheduledTask -TaskName "AICyberAuditBox_Startup"
```

**Manual Clean Stop (stops all servers and docker DB):**

```
  taskkill /F /IM llama-server.exe /T
taskkill /F /IM streamlit.exe /T
cd C:\Users\veeresh988V\Desktop\llama\AICyberSecurityAuditBoxV
docker-compose down
```

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**Troubleshooting Notes:**
- Connection Timeout: Verify that Azure NSG port 8501 rule is active.
- LLM Server fails: Verify GGUF files exist in root folder and model name is correct.
- Database fails: Open Docker Desktop and verify that the WSL2 subsystem is active.